Subhrajit Mishra

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Background

Sep 2026 **Master of Science**, Computer Science and Engineering, UC San Diego, USA — *Incoming fall'26*

May 2024 **Bachelor of Technology**, Department of Electrical Engineering, IIT Kanpur, India — *CPI: 9.2/10*

May 2020 **Grade XII**, Yogiraj Public School, Kota, India — *Percentage: 95.0%*

Research Interests

Efficient Deep Learning · Medical Imaging · Computer Vision · Multimodal Learning · Image and Video Signal Processing

Research Experience

Multimodal Contrastive Learning - Undergraduate Project

IIT Kanpur | Code

Supervisor: Dr. Tushar Sandhan – *Perception and Intelligence Lab, IIT Kanpur* — *Jan - Mar 2024*

- Aim: Diagnosis of **Myocardial infarction** and **Coronary artery disease** using tabular and imaging patient data
- Designed a generalizable **SimCLR** based **contrastive learning multimodal** framework with **InfoNCE** loss
- Generalized the framework to **Medical equipment classification** for FDA/CDSCO using images and description
- Created an unimodal model for equipment classification with tabular data using **BERT** to compare performance

AI-based In-Loop Filters in post VVC

Samsung Research

- Design low-complex **hardware-integrable** neural network for suppressing compression and banding **artifacts**
- Combined artifact-aware **handcrafted filters** with custom designed **attention**, and convolutional core block
- Delivered state-of-the-art Bjøntegaard Delta Bitrate (BD-BR) gains at **ultra-low complexity** less than 500FLOPs/pix
- Integrated the model with the VVC Test Model, with **dedicated signalling** to operate in parallel to Deblocking filter

Neural Network based Intra Prediction

Samsung Research

- Aim: Construct **unified** NN-based intra-frame prediction handling **varying input** and output Coding unit sizes
- Designed an **adaptable** and **configurable** CNN based network with an intelligent pre- and post-processing stages
- Integrated the NN-based intra-prediction mode **alongside** classical predictors in post-VVC, enabling C++ inference
- Replaced seven separate post-VVC model variants and achieved **State of Art** BD-BR gains at 1KFlops/pix complexity

Unmanned Aerial Vehicle (UAV) guided Unmanned Ground Vehicle (UGV) navigation

IIT Kanpur | Code

InterIIT TechMeet 10.0, DRDO challenge, *Problem statement and solution documentation* — *Feb - Mar 2024*

- **Autonomous traversal:** Incorporated a **PID velocity Controller** for modulating speed and a **Pure Pursuit** Controller for steering in the UGV for traversal on mountainous terrains, using odometry obtained by a hovering UAV
- **Real-time motion planning:** UAV camera image **segmentation** to identify road, the center and orientation of UGV
- **Autonomous takeoff of UAV and mean path detection for UGV:** Ensured minimal derouting from mean path

Open Advanced Professional Video (APV) Codec

Samsung Research

- Contributed core algorithms for APV, an open-source, **perceptually lossless** professional-grade video codec
- Optimized codec encoding and decoding speed by using **platform-specific** intrinsic tools (**NEON** and **AVX**)
- Integrated APV codec on Samsung devices and Premiere Pro for **National Association of Broadcasters**(NAB) demo

Patents

Methods and system for AI-based pixel-block prediction for video codec

Filed

A unified, **hardware-efficient** network for intra prediction using reference lines to predict coding units of **variable sizes**.

NN-Based Reference Sample Enhancement for Intra Prediction

Filed

An **enhancement network** to improve the input to intra prediction by enhancing the already decoded reference pixels.

Adaptive Visual Artifact Removal in Video Codec using AI Inloop Filters

Filed

Content-based partitioning of a Mixture of Experts for NN - loop filtering to improve subjective and objective **quality**.

Inverted-L-Scan Method for AI-based Intra-frame prediction in video codec

Filed

A **pre- and post-processing** method to prepare neighboring pixels for a **unified** NN-based intra prediction model.

Dynamic AI-based multiplexing for combining NN and classical Loop Filters Filed
Scalar and patch-level multiplexing of existing loop filters with NN-based loop filter to remove compression artifacts.

On-the-Fly Kernel Adaptation using Affine Transform in AI Loop Filtering Filed
Using the Quantization Parameter as a prior for routing in the neural network and dynamically updating the weights.

Methods & Systems for performing Adaptable Multi-filter refinement Loop Filtering Filed
Creating a learned combination of relevant classical filters and a trainable filter to generate an Edge-Refinement block.

Dynamic Local patch modulation in AI-based In-loop Filtering in Video Codec Filed
A spatial attention-inspired configuration to provide learned weightage to the local neighborhood of a patch.

Academic Projects

Atari Games with Reinforcement Learning IIT Kanpur | Code
Supervisor: Dr. Subrahmanya Swamy Peruru – *Department of Electrical Engineering, IIT Kanpur* *Mar - Apr 2024*

- Created a **DQN** and a **DDQN** model of five layers with the **cart pole environment** for playing **Atari games**
- **Results:** Model was run for **Breakout & Pong** games; it beat the baseline agent reaching max reward convergence
- Added **LSTM** for **DRQN** model, matching published results for **partial observability** in **flickering** environment

Antenna Designing for Communication Systems IIT Kanpur | Code
Supervisor: Dr. Raghvendra Kumar Chaudhary – *Radio Frequency, mmWave Lab, IIT Kanpur* *Mar - May 2024*

- Produced mmWave **horn antennas** for 75-110GHz, with **frequency selective surfaces** to control cross polarization
- Simulated designs of various types of antennas, finding optimal structures for best gains, **directivity** and **S plots**
- **Impact:** Designed **X-Band Mono-pulse** Conical Feed Antenna for **Low Earth Orbit** (LEO) Satellite Ground Station

Machine Learning Course Project IIT Kanpur | Code
Supervisor: Dr. Purushottam Kar – *Department of Computer Science Engineering, IIT Kanpur* *Jan - Apr 2023*

- Created **linear models** to crack simple & advanced **XORRO PUF** with >99% accuracy at ultra-low complexity
- Developed a Decision tree based model to play a hybrid of **Hangman** and **Wordle** with average query rate of 3.99
- Created an **electrochemical air quality sensor caliberator** using **Elastic Net** and **Extra tree regressor**

Professional Experience

Samsung Research | Advanced Engineer Bengaluru, India
Next Generation Video Compression Technology (Full-Time) *July 2024 - August 2026*

- **Neural Network Video Codec Development for Upcoming standard H.267:** Development and integration of NN-based algorithms for In-loop filters and Intra Prediction over existing standard, H.266 - Versatile Video Codec
- **Advanced Professional Video:** Developing perceptually lossless codec for **ultrasound probe** and pro video editor
- **MPEG-5: EVC and FFMPEG:** Open source codec development, core intrinsic optimization and on-device integration

EVERSANA: Asia-Pacific Mumbai, India
Digital and Data solutions (Internship) *May 2023 - July 2023*

- Conceptualization of **Willingness to Recommend** (WTR) and **Intent to pay** (ITP) for medical consumables
- Uncovering adequate **clusters** of physicians and hospitals as per behavioral implications and habitual practices
- Identifying **optimal interventions** for cohorts and creating a software of the solution to be used by the client

Honors and Awards

Jan 2025 **AI expert and Software Pro** for exceptional deep learning and software development skills *Samsung*
Apr 2024 **Academic Excellence Award** for exceptional performance in the year 2021, 2023, & 2024 *IIT Kanpur*
Mar 2022 **InterIIT Techmeet 10, Silver Medal** Aerial Vehicle guided Ground vehicle navigation *IIT Kanpur*
2018 - '20 **KVPY and NTSE Scholarships**, Ministry of Science & Technology, Govt. of India *India*

Skills

Programming: Python, C++, C, MATLAB, HTML/CSS, Solidity, Object-oriented programming, MySQL
ML/DL: Pytorch, Tensorflow, Scikit-Learn, OpenCV, Numpy, Pandas, LaTeX
Systems: Linux, Git, bash, Neon/AVX intrinsics, ROS, Gazebo, Visual Studio

Relevant Coursework

Artificial Intelligence	Advanced Machine Learning, Deep Learning specialization, Reinforcement Learning, Convolutional Neural Networks, Computer Vision, Sequence Models
Computer Science	Intro to programming, Data structures and Algorithms, Computer Organization, Operating System, Computer Networking, Database management system, Cyber Security
Mathematics	Real Analysis, Complex Analysis, Linear Algebra, Ordinary and Partial Differential Equations, Probability and Statistics, Stochastic Process
Electrical Engineering	Signal Processing, Image Processing, Digital Electronics, Microelectronics, Principles of Communications, Electromagnetic theory, Electrical Engineering Labs - I & II

Test Scores

Aug 2025 **TOEFL iBT** Score: 112/120 *ETS*

Sept 2025 **GRE General Test** Score: 325/340 (Breakup - Quant: 170/170; Verbal: 155/170) *ETS*

Teaching Experience

Team Head — Aerial Robotics, IIT Kanpur
Aug ’22 - Nov ’22

- Recruited **25 Junior members** for my team with induction to the required tech stack and simulation environments
- Mentored 7 junior members in a **mini-project** on **swarm robotics** guiding them in drone flight control & automation

Project Guide — Stamatics Society, IIT Kanpur
May ’23 - Aug ’23

- Guided around **98 Undergrads** to master **Image processing algorithms** and utilizing them for computer vision
- Spearheaded **diverse projects** curated to depict the effect of classical image processing in Deep learning

Academic Mentor — Counselling Service, IIT Kanpur
Aug ’21 - Mar ’22

- Conducted various **academic classes** for the course PHY102, teaching various topics to around **600 students**
- Personally guided **20+ freshman students** throughout their **first year** at all **academic fronts**

Positions of Responsibility

Team Head — Aerial Robotics, IIT Kanpur
May ’21 - Aug ’23

- Learnt **robot programming** and Development of **drones** with simulation environments and hardware testing
- **Represented IIT Kanpur** at various national and international competitions including **InterIIT Techmeets**
- Led a two-tier team of **30 members** enabling **participation & sponsorships** for several **Aerial Robotics** events

Secretary — Entrepreneurship Cell, IIT Kanpur
Apr ’21 - May ’22

- Spearheaded a three tier team of **100 students** for the successful conduction of **Entrepreneurship Summit**, 2022
- **Finalized sponsorship deals** and signed MoUs with various companies and continued as their first point of contact

Volunteering and Extra-curriculars

- Spearheaded various **state level science exhibitions** and **Community Service Camps** as a commanding member of the civil defence service unit, **National Service Scheme** (NSS)
- Have been a regular **blood donor** with active involvement in organizing blood donation camps, since Oct’ 21
- Active participant in **Competitive Programming** solving **700+** problems with a maximum rating of **1703**
- Member of **Debating Society and Bicycling Society, IIT Kanpur**, winning various **inter-collegiate** competitions
- Victoriously led the **Music, Football and Quizzing teams** in several **inter-college** and **inter-school** events